NATIONAL SENIOR CERTIFICATE

GRADE 12

MATHEMATICS P1

NOVEMBER 2024

MARKS: 150 TIME: 3 hours

This question paper consists of 3 pages.

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QUESTION 1

1.1 Solve for x:

1.1.1 x^2 - 5x + 6 = 0 (3)

1.1.2 2x^2 + x - 4 = 0 (correct to TWO decimal places) (4)

1.2 Solve simultaneously for x and y: y = 2x - 1 and x^2 + y^2 = 13 (6)

1.3 Determine the value of k if the roots of x^2 + kx + 9 = 0 are equal. (3)

[16]

QUESTION 2

Given the arithmetic sequence: 5 ; 9 ; 13 ; ...

2.1 Determine the general term Tn. (2)

2.2 Calculate the sum of the first 30 terms. Show all working and explain each step of your calculation clearly, using the appropriate formula from the information sheet.

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2.2 continued: hence determine which term equals 121. (4)

[6]

QUESTION 3

3.1 Simplify: (2^(x+2) - 2^x) / (3 . 2^x) (3)

3.2 Prove that sqrt(50) - sqrt(18) = 2 sqrt(2) (3)

[6]

TOTAL: 150